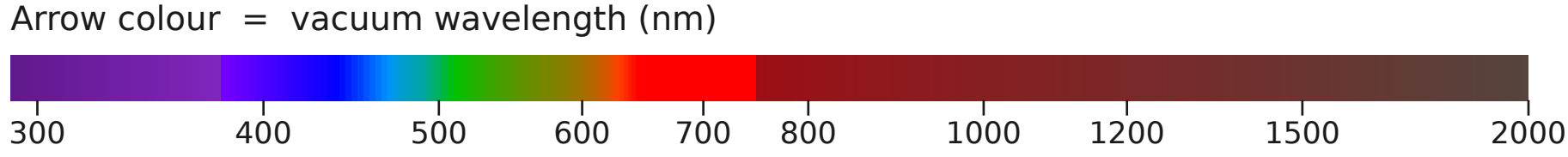


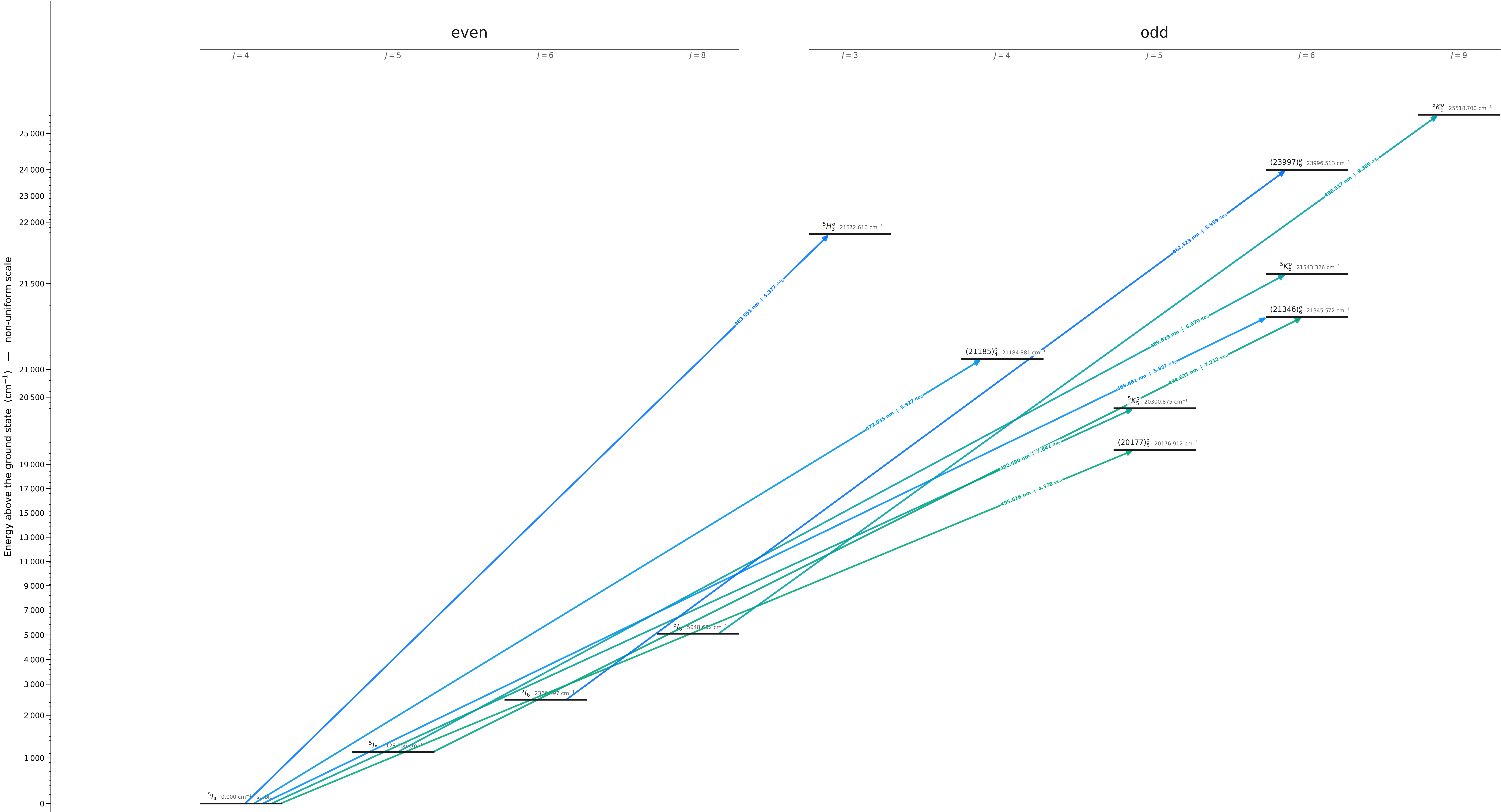
Neodymium

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J || e r || J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Nd (Nd I). Strongest classified transitions below 2 μm (NIST): 12 levels, 9 lines.

Nd⁺ ionisation limit 44560.11 cm^{-1} = 5.52475 eV (224.416 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ (ea_0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^5I_6 - (23997)^5_6$	462.3226	462.1932	648.4486	5.959	5.600e+07	-	-	-
$^5I_4 - ^5H^3_3$	463.5508	463.4210	646.7306	5.377	8.400e+07	-	-	-
$^5I_4 - (21346)^5_6$	468.4812	468.3501	639.9242	5.857	5.200e+07	-	-	-
$^5I_4 - (21185)^5_4$	472.0347	471.9027	635.1068	3.927	3.300e+07	-	-	-
$^5I_8 - ^5K^2_9$	488.5174	488.3810	613.6781	9.809	8.800e+07	-	-	-
$^5I_5 - ^5K^2_8$	489.8294	489.6927	612.0344	6.670	5.900e+07	-	-	-
$^5I_4 - ^5K^2_5$	492.5896	492.4521	608.6049	7.642	9.000e+07	-	-	-
$^5I_5 - (21346)^5_6$	494.6206	494.4826	606.1059	7.212	6.700e+07	-	-	-
$^5I_4 - (20177)^5_5$	495.6160	495.4777	604.8886	4.378	2.900e+07	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

¹⁴³Nd hyperfine structure ($I = 7/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)							
5I_4	-195.649(9)	122.25(28)	F=1/2: +3483.89	F=3/2: +3174.98	F=5/2: +2663.25	F=7/2: +1953.37	F=9/2: +1051.90	F=11/2: -32.75	F=13/2: -1290.27	F=15/2: -2708.52

Measured A, B constants; sources in the data file.